

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Piping Mechanical MTO
Document Number	EA-BFPCL-GPMC-GGPL-FEED-MPP-LST-015
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Piping Mechanical MTO

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Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	03.08.22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-MPP-LIST-015

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES

Document No.	Description

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1. MATERIAL TAKE OFF FOR PIPING BULKS - BBOU MANIFOLD A1-N4														COMMENTS
SINO	ITEM DESCRIPTION	MATERIAL CLASS	SIZE 1	SIZE 2	SCHEDULE / THICKNESS 1	SCHEDULE / THICKNESS 2	PRESSURE RATING	DESIGN CODE	MATERIAL DESCRIPTION	UNIT	QUANTITY REQUIRED	CONTINGENCY	QUANTITY TO PROCURE	
PIPES														
1	PIPE	C70	24"	--	24.2mm	--	900#	ASME B36.10/B36.19,	SEAMLESS/WELDED, 100% RADIOGRAPHY, ASME B36.10/B36.19, ASTM A790 S31803/ASTM A928 S31803 CL 11 (12m SRL)	m	16.9	7.1	24	Material is captured in LLI List
2	PIPE	C70	2"	--	80S	--	900#	ASME B36.10/B36.19,	SEAMLESS/WELDED, 100% RADIOGRAPHY, ASME B36.10/B36.19, ASTM A790 S31803/ASTM A928 S31803 CL 1 (6m SRL)	m	13.6	4.4	18	Material is captured in LLI List
3	PIPE	C03	24"	--	100	--	900#	ASME B36.10M	WELDED, SAW 100% RADIOGRAPHY, ASME B36.10, API SPEC 5L GRADE X60 PSL 2 (12m SRL)	m	70.5	1.5	72	Material is captured in LLI List
4	PIPE	C03	8"	--	120	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	30.2	5.8	36	Material is captured in LLI List
5	PIPE	C03	6"	--	120	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	29.3	0.7	30	Material is captured in LLI List
6	PIPE	C03	4"	--	120	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	54.9	5.1	60	Material is captured in LLI List
7	PIPE	C03	2"	--	160	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	36.4	5.6	42	Material is captured in LLI List
8	PIPE	C03	3/4"	--	160	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (6m SRL)	m	0.2	5.8	6	Material is captured in LLI List
9	PIPE	B03	6"	--	STD	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (12m SRL)	m	22.7	1.3	24	Material is captured in LLI List
10	PIPE	B03	4"	--	STD	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (12m SRL)	m	25.4	4.6	30	Material is captured in LLI List
11	PIPE	B03	2"	--	XS	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (12m SRL)	m	1.3	4.7	6	Material is captured in LLI List
12	PIPE	B03	3/4"	--	XS	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (12m SRL)	m	0.2	5.8	6	Material is captured in LLI List
TEES														
13	EQUAL TEE	C03	24"	24"	100	100	900#	ASME B16.9	ASTM A860 WPHY 60, BW, ASME B16.9	PCS	8	1	9	
14	REDUCING TEE	C03	8"	4"	120	120	900#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	2	0	2	
15	EQUAL TEE	C03	6"	6"	120	120	900#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	1	0	1	
16	EQUAL TEE	C03	4"	4"	120	120	900#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	5	1	6	
17	EQUAL TEE	C03	2"	2"	160	160	900#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	3	0	3	
18	EQUAL TEE	B03	6"	6"	STD	STD	150#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	4	0	4	
19	REDUCING TEE	B03	6"	4"	STD	STD	150#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	3	0	3	
WELDOLETS														
20	WELDOLET	C70	24"	8"	24.2mm	24.2mm	900#	MSS-SP-97	WELDOLET, MSS SP-97, ASTM A182 F51	PCS	3	0	3	
21	WELDOLET	C70	24"	2"	24.2mm	80S	900#	MSS-SP-95	WELDOLET, MSS SP-97, ASTM A182 F51	PCS	1	0	1	
22	WELDOLET	C03	24"	8"	100	120	900#	MSS-SP-96	WELDOLET, MSS SP-97, ASTM A105N	PCS	2	0	2	
23	WELDOLET	C03	24"	6"	100	120	900#	MSS-SP-97	WELDOLET, MSS SP-97, ASTM A105N	PCS	1	0	1	
24	WELDOLET	C03	24"	4"	100	120	900#	MSS-SP-97	WELDOLET, MSS SP-97, ASTM A105N	PCS	1	0	1	
25	WELDOLET	C03	24"	2"	100	STD	900#	MSS-SP-97	WELDOLET, MSS SP-97, ASTM A105N	PCS	15	2	17	
26	WELDOLET	C03	8"	2"	120	STD	900#	MSS-SP-97	WELDOLET, MSS SP-97, ASTM A105N	PCS	4	0	4	
SOCKOLETS														
27	SOCKOLET	C03	24"	1 1/2"	100	160	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	2	0	2	
28	SOCKOLET	C03	24"	3/4"	100	160	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	4	0	4	
29	SOCKOLET	C03	8"	3/4"	120	160	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	1	0	1	
30	SOCKOLET	C03	6"	3/4"	120	160	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	2	0	2	
31	SOCKOLET	C03	4"	3/4"	120	160	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	4	0	4	
32	SOCKOLET	B03	6"	3/4"	STD	160	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	4	0	4	

REDUCERS														
33	CONCENTRIC REDUCER	C03	6"	4"	140	XXS	900#	ASME B16.8	ASTM A234 GR. WPB; SMLS; BW; ASME B16.9	PCS	2	0	2	
ELBOW														
34	90 DEG. LR. ELBOW	C70	24"	--	24.2mm	--	1500#	ASME B16.9	ASTM A815 S31803 WP-SWP-WX; ASME B16.9	PCS	1	0	1	
35	90 DEG. LR. ELBOW	C70	2"	--	80S	--	1500#	ASME B16.9	ASTM A815 S31803 WP-SWP-WX; ASME B16.9	PCS	5	1	6	
36	90 DEG. LR. ELBOW	C03	24"	--	100	--	900#	ASME B16.9	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	12	1	13	
37	90 DEG. LR. ELBOW	C03	8"	--	120	--	900#	ASME B16.9	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	14	1	15	
38	90 DEG. LR. ELBOW	C03	6"	--	120	--	900#	ASME B16.9	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	6	1	7	
39	90 DEG. LR. ELBOW	C03	4"	--	120	--	900#	ASME B16.9	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	33	3	36	
40	90 DEG. LR. ELBOW	C03	2"	--	160	--	900#	ASME B16.9	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	36	4	40	
41	90 DEG. LR. ELBOW	B03	6"	--	STD	--	150#	ASME B16.9	ASTM A234 GR. WPB; SMLS; BW; ASME B16.9	PCS	9	1	10	
42	90 DEG. LR. ELBOW	B03	4"	--	STD	--	150#	ASME B16.10	ASTM A234 GR. WPB; SMLS; BW; ASME B16.9	PCS	9	1	10	
FLANGES														
43	FLANGE	C70	24"	--	24.2mm	--	900#	ASME B16.5	ASTM A182 F51, 'WN, 1500#, RTJ, ASME B16.47 SERIES A	PCS	5	1	6	
44	BLIND FLANGE	C70	24"	--	24.2mm	--	900#	ASME B16.5	ASTM A182 F51, 1500#, RTJ, ASME B16.47 SERIES A	PCS	1	0	1	
45	FLANGE	C03	24"	--	100	--	900#	ASME B16.5	ASTM A694 F60, 'WN, 900#, RTJ, ASME B16.47 SERIES A	PCS	21	2	23	
46	BLIND FLANGE	C03	24"	--	100	--	900#	ASME B16.5	ASTM A694 F60, 900#, RTJ, ASME B16.47 SERIES A	PCS	1	0	1	
47	FLANGE	C03	8"	--	140	--	900#	ASME B16.5	ASTM A105N; WN; 900#; RTJ; ASME B16.5	PCS	6	1	7	
48	FLANGE	C03	6"	--	140	--	900#	ASME B16.5	ASTM A105N; WN; 900#; RTJ; ASME B16.5	PCS	8	1	9	
49	FLANGE	C03	4"	-	140	-	900#	ASME B16.5	ASTM A105N; WN; 900#; RTJ; ASME B16.5	PCS	32	3	35	
50	FLANGE	C03	2"	--	160	--	900#	ASME B16.5	ASTM A105N; WN; 900#; RTJ; ASME B16.5	PCS	46	5	51	
51	FLANGE	C70	2"	--	80S	--	900#	ASME B16.5	ASTM A105N; WN; 900#; RTJ; ASME B16.5	PCS	3	0	3	
52	FLANGE	C03	3/4"	--	160	--	1500#	ASME B16.5	ASTM A105N; SW; 1500#; RTJ; ASME B16.5	PCS	10	1	11	
53	BLIND FLANGE	C03	3/4"	--	160	--	1500#	ASME B16.5	ASTM A105N; SW; 1500#; RTJ; ASME B16.5	PCS	10	1	11	
54	FLANGE	B03	6"	--	STD	--	150#	ASME B16.5	ASTM A105N; WN; 150#; RF; ASME B16.5	PCS	16	2	18	
55	BLIND FLANGE	B03	6"	--	STD	--	150#	ASME B16.5	ASTM A105N; WN; 150#; RF; ASME B16.5	PCS	1	0	1	
56	FLANGE	B03	4"	--	STD	--	150#	ASME B16.5	ASTM A105N; WN; 150#; RF; ASME B16.5	PCS	5	1	6	
57	BLIND FLANGE	B04	2"	--	STD	--	150#	ASME B16.6	ASTM A105N; WN; 150#; RF; ASME B16.6	PCS	2	0	2	
58	FLANGE	B03	3/4"	--	XXS	--	150#	ASME B16.5	ASTM A105N; SW; 150#; RF; ASME B16.5	PCS	4	0	4	
59	BLIND FLANGE	B03	3/4"	--	XXS	--	150#	ASME B16.5	ASTM A105N; SW; 150#; RF; ASME B16.5	PCS	4	0	4	
SPECTACLE BLINDS														
60	SPECTACLE BLIND	C03	4"	--	--	--	900#	ASME B16.48	ASTM A516 GR.70; 900#; ASME B16.48	PCS	1	0	1	

GASKETS														
61	GASKET	C70	24"	--	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, CL1500, ASME B16.20, TYPE 316	PCS	3	3	6	
62	GASKET	C70	2"	--	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, CL1500, ASME B16.20, TYPE 317	PCS	3	3	6	
63	GASKET	C03	24"	--	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	20	20	40	
64	GASKET	C03	8"	--	--	--	900#	'ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	8	8	16	
65	GASKET	C03	6"	--	--	--	900#	'ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	8	8	16	
66	GASKET	C03	4"	--	--	--	900#	'ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	35	35	70	
67	GASKET	C03	2"	--	--	--	900#	'ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	50	50	100	
68	GASKET	C03	1 1/2"	--	--	--	900#	'ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	2	2	4	
69	GASKET	C03	3/4"	--	--	--	900#	'ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	10	10	20	
70	GASKET	B03	6"	--	--	--	150#	ASME B16.20	SPIRAL WOUND, LOW STRESS, 150#, ASME B16.20, SS 316 WINDING 4.5MM THICK, GRAPHITE FILLER, INNER AND CENTERING RING 3.2 MM THICK	PCS	17	17	34	
71	GASKET	B03	4"	--	--	--	150#	ASME B16.20	SPIRAL WOUND, LOW STRESS, 150#, ASME B16.20, SS 316 WINDING 4.5MM THICK, GRAPHITE FILLER, INNER AND CENTERING RING 3.2 MM THICK	PCS	5	5	10	
72	GASKET	B03	2"	--	--	--	150#	ASME B16.20	SPIRAL WOUND, LOW STRESS, 150#, ASME B16.20, SS 316 WINDING 4.5MM THICK, GRAPHITE FILLER, INNER AND CENTERING RING 3.2 MM THICK	PCS	8	8	16	
73	GASKET	B03	1 1/2"	--	--	--	150#	ASME B16.20	SPIRAL WOUND, LOW STRESS, 150#, ASME B16.20, SS 316 WINDING 4.5MM THICK, GRAPHITE FILLER, INNER AND CENTERING RING 3.2 MM THICK	PCS	1	1	2	
74	GASKET	B03	3/4"	--	--	--	150#	ASME B16.20	SPIRAL WOUND, LOW STRESS, 150#, ASME B16.20, SS 316 WINDING 4.5MM THICK, GRAPHITE FILLER, INNER AND CENTERING RING 3.2 MM THICK	PCS	5	5	10	
STUDBOLTS & NUTS														
75	STUD BOLTS & HEX NUTS	C70	2+1/2"	605mm	--	--	1500#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A453 GRADE 660 CLASS B	PCS	60	12	72	
76	STUD BOLTS & HEX NUTS	C70	7/8"	160mm	--	--	1500#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A453 GRADE 660 CLASS B	PCS	24	5	29	
77	STUD BOLTS & HEX NUTS	C03	2+1/2"	505mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	400	80	480	
78	STUD BOLTS & HEX NUTS	C03	1+3/8"	230mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	96	19	115	
79	STUD BOLTS & HEX NUTS	C03	1+1/8"	215mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	96	19	115	
80	STUD BOLTS & HEX NUTS	C03	1+1/8"	190mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	280	56	336	
81	STUD BOLTS & HEX NUTS	C03	7/8"	160mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	400	80	480	
82	STUD BOLTS & HEX NUTS	C03	1"	155mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	8	2	10	
83	STUD BOLTS & HEX NUTS	C03	3/4"	125mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	40	8	48	
84	STUD BOLTS & HEX NUTS	B03	3/4"	105mm	--	--	150#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	136	27	163	
85	STUD BOLTS & HEX NUTS	B03	5/8"	95mm	--	--	150#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	40	-	#VALUE!	
86	STUD BOLTS & HEX NUTS	B03	5/8"	85mm	--	--	150#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A154	PCS	32	6	38	
87	STUD BOLTS & HEX NUTS	B03	1/2"	75mm	--	--	150#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A152	PCS	4			
88	STUD BOLTS & HEX NUTS	B03	1/2"	65mm	--	--	150#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	20	4	24	
SPECIAL PIPING ITEMS														
89	QUICK DISCONNECT COUPLING	C03	2"	--	--	--	900#	ASTM F1122	BODY: ASTM A105N; CAP: ASTM A234 GR. WPB; HOSE: NITRILE / EPDM RUBBER WITH HT SYNTHETIC REINFORCEMENT; FLANGE: ASTM A105N; SEAL: RPTFE	1 SET	3	0	3	

2.0 MATERIAL TAKE OFF FOR PIPING VALVES BBOU MANIFOLD														COMMENTS
S/NO	SIZE	MATERIAL CLASS	VALVE TYPE	VALVE CODE	BORE TYPE	END CONNECTION	PRESSURE CLASS	DESIG CODE	ITEM & MATERIAL DESCRIPTION	UNIT	REQUIRED QUANTITY	CONTINGENCY	TOTAL QUANTITY TO	
1	24" SDV 1001	C03	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1	Material is captured in LLI List
1	24"	C03	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	4	0	4	Material is captured in LLI List
1	24"	C03	BALL	—	FB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	2	0	2	Material is captured in LLI List
2	8"	C03	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	4	0	4	Material is captured in LLI List
3	6"	C03	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	4	0	4	Material is captured in LLI List
4	6"	B03	BALL	—	RB	RF FLANGED WITH SMOOTH FINISH	150#		FLGD, RF, CL150, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	6	0	6	Material is captured in LLI List
5	4" BDV 1001	C03	BALL	—	FB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1	Material is captured in LLI List
5	4"	C03	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	13	0	13	Material is captured in LLI List
6	4"	B03	BALL	—	RB	RF FLANGED WITH SMOOTH FINISH	150#		FLGD, RF, CL150, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	1	0	1	Material is captured in LLI List
6	2"	D70	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	1500#		FLGD, RJ, CL1500, API SPEC 6D, ASTM A182 F51 OR ASTM A995 GR 4A BODY, UNS S31803 TRIM, SPB/TPE, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1	Material is captured in LLI List
7	2"	C03	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	21	0	21	Material is captured in LLI List
8	2"	B03	BALL	—	RB	RF FLANGED WITH SMOOTH FINISH	150#		FLGD, RF, CL150, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	4	0	4	Material is captured in LLI List
9	3/4"	C03	BALL	—		SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	1500#		SW, CL1500, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	10	0	10	Material is captured in LLI List
10	3/4"	B03	BALL	—		SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	800#		SW, CL800, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	4	0	4	Material is captured in LLI List
11	4"	C03	GLOBE	—		RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SWIVEL PLUG DISC, HNDWHL OPR W/ GEARBOX	PCS	4	0	4	Material is captured in LLI List
12	2"	CO3	GLOBE	—		RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SWIVEL PLUG DISC, HNDWHL OPR W/ GEARBOX	PCS	7	0	7	Material is captured in LLI List

1. MATERIAL TAKE OFF FOR PIPING BULKS - ELEPA MANIFOLD														COMMENTS
S/NO	ITEM DESCRIPTION	MATERIAL CLASS	SIZE 1	SIZE 2	SCHEDULE / THICKNESS 1	SCHEDULE / THICKNESS 2	PRESSURE RATING	DESIGN CODE	MATERIAL DESCRIPTION	UNIT	QUANTITY REQUIRED	CONTINGENCY	QUANTITY TO PROCURE	
PIPES														
1	PIPE	D70	36"	--	59.2mm	--	1500#	ASME B36.10/B36.19,	SEAMLESS/WELDED, 100% RADIOGRAPHY, ASME B36.10/B36.19, ASTM A790 S31803/ASTM A928 S31803 CL 1 (12m SRL)	m	13.5	10.5	24	Material is captured in LLI List
2	PIPE	D06	36"	--	76.1mm	--	1500#	ASME B36.10/B36.19,	WELDED, SAW 100% RADIOGRAPHY, ASME B36.10, API SPEC 5L GRADE X60 PSL 2 (12m SRL)	m	4.0	8.0	12	Material is captured in LLI List
3	PIPE	D06	2"	--	11.07mm	--	1500#	ASME B36.10/B36.19,	SEAMLESS/WELDED, 100% RADIOGRAPHY, ASME B36.10/B36.19, ASTM A790 S31803/ASTM A928 S31803 CL 1 (12m SRL)	m	13.6	4.4	18	Material is captured in LLI List
4	PIPE	C06	36"	--	38.2mm	--	900#	ASME B36.10M	WELDED, SAW 100% RADIOGRAPHY, ASME B36.10, API SPEC 5L GRADE X60 PSL 2 (12m SRL)	m	62.1	9.9	72	Material is captured in LLI List
5	PIPE	C06	24"	--	100	--	900#	ASME B36.10M	WELDED, SAW 100% RADIOGRAPHY, ASME B36.10, API SPEC 5L GRADE X60 PSL 2 (12m SRL)	m	18.9	5.1	24	Material is captured in LLI List
6	PIPE	C06	12"	--	140	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	31.4	4.6	36	Material is captured in LLI List
7	PIPE	C06	8"	--	140	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	29.1	6.9	36	Material is captured in LLI List
8	PIPE	C06	6"	--	160	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	135.2	2.8	138	Material is captured in LLI List
9	PIPE	C06	4"	--	XXS	--	900#	ASME B36.10M	*SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	71.6	0.4	72	Material is captured in LLI List
10	PIPE	C06	2"	--	XXS	--	900#	ASME B36.10M	*SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	28.6	1.4	30	Material is captured in LLI List
11	PIPE	C06	3/4"	--	XXS	--	900#	ASME B36.10M	*SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (6m SRL)	m	1.0	5.0	6	Material is captured in LLI List
12	PIPE	B06	6"	--	XS	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (12m SRL)	m	39.2	2.8	42	Material is captured in LLI List
13	PIPE	B06	4"	--	120	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (12m SRL)	m	47.0	1.0	48	Material is captured in LLI List
14	PIPE	B06	2"	--	XXS	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (12m SRL)	m	1.3	4.7	6	Material is captured in LLI List
15	PIPE	B06	3/4"	--	XXS	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (6m SRL)	m	1.0	5.0	6	Material is captured in LLI List
TEES														
16	EQUAL TEE	C06	36"	36"	38.2mm	38.2mm	900#	ASME B16.9	ASTM A860 WPHY 60, BW, ASME B16.9	PCS	4	0	4	
17	REDUCING TEE	C06	36"	24"	38.2mm	100	900#	ASME B16.9	ASTM A860 WPHY 60, BW, ASME B16.9	PCS	1	0	1	
18	REDUCING TEE	C06	36"	16"	38.2mm	140	900#	ASME B16.9	ASTM A860 WPHY 60, BW, ASME B16.9	PCS	2	0	2	
19	EQUAL TEE	C06	24"	24"	100	100	900#	ASME B16.9	ASTM A860 WPHY 60, BW, ASME B16.9	PCS	0	0	0	
20	REDUCING TEE	C06	8"	4"	140	XXS	900#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	1	0	1	
21	EQUAL TEE	C06	6"	6"	160	160	900#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	1	0	1	
22	REDUCING TEE	C06	6"	4"	160	XXS	900#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	2	0	2	
23	EQUAL TEE	C06	4"	4"	XXS	XXS	900#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	5	1	6	
24	EQUAL TEE	C06	2"	2"	XXS	XXS	900#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	2	0	2	
25	EQUAL TEE	B06	6"	6"	XS	XS	150#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	5	1	6	
26	REDUCING TEE	B06	6"	4"	XS	140	150#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	4	0	4	
WELDOLETS														
27	WELDOLET	D70	36"	8"	59.2mm	14.9mm	1500#	MSS-SP-97	WELDOLET, MSS SP-97, ASTM A182 F51	PCS	3	0	3	
28	WELDOLET	D70	36"	2"	59.2mm	80S	1500#	MSS-SP-98	WELDOLET, MSS SP-97, ASTM A182 F51	PCS	1	0	1	
29	WELDOLET	C06	36"	6"	38.2mm	160	900#	MSS-SP-97	WELDOLET, MSS SP-97, ASTM A105N	PCS	1	0	1	
30	WELDOLET	C06	36"	4"	38.2mm	XXS	900#	MSS-SP-97	WELDOLET, MSS SP-97, ASTM A105N	PCS	1	0	1	
31	WELDOLET	C06	36"	2"	38.2mm	XXS	900#	MSS-SP-97	WELDOLET, MSS SP-97, ASTM A105N	PCS	8	1	9	
32	WELDOLET	C06	24"	8"	100	140	900#	MSS-SP-97	*WELDOLET, MSS SP-97, ASTM A105N	PCS	1	0	1	
33	WELDOLET	C06	24"	2"	100	XXS	900#	MSS-SP-97	*WELDOLET, MSS SP-97, ASTM A105N	PCS	2	0	2	
34	WELDOLET	C06	12"	4"	140	XXS	900#	MSS-SP-97	*WELDOLET, MSS SP-97, ASTM A105N	PCS	2	0	2	
35	WELDOLET	C06	12"	2"	140	XXS	900#	MSS-SP-97	*WELDOLET, MSS SP-97, ASTM A105N	PCS	4	0	4	
36	WELDOLET	C06	8"	2"	140	XXS	900#	MSS-SP-97	*WELDOLET, MSS SP-97, ASTM A105N	PCS	2	0	2	
37	WELDOLET	C06	12"	4"	80	80	600#	MSS-SP-97	ASTM A105N; BW; MSS-SP-97	PCS	1	0	1	

SOCKOLETS														
38	SOCKOLET	C06	36"	1 1/2"	38.2mm	XXS	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	2	0	2	
39	SOCKOLET	C06	36"	3/4"	38.2mm	XXS	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	2	0	2	
40	SOCKOLET	C06	24"	3/4"	100	XXS	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	1	0	1	
41	SOCKOLET	C06	12"	3/4"	140	XXS	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	4	0	4	
42	SOCKOLET	C06	8"	3/4"	140	XXS	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	1	0	1	
43	SOCKOLET	C06	6"	3/4"	160	XXS	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	5	1	6	
44	SOCKOLET	C06	4"	3/4"	XXS	XXS	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	1	0	1	
45	SOCKOLET	B06	6"	3/4"	XS	XXS	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	6	1	7	
REDUCERS														
46	CONCENTRIC REDUCER	C06	16"	12"	140	140	900#	ASME B16.9	ASTM A234 WPB, BW, WPBASME B16.9,	PCS	2	0	2	
47	CONCENTRIC REDUCER	C06	6"	4"	160	XXS	900#	ASME B16.8	ASTM A234 GR. WPB; SMLS; BW; ASME B16.9	PCS	4	0	4	
ELBOW														
48	90 DEG. LR. ELBOW	D07	36"	--	59.2mm	--	1500#	ASME B16.9	ASTM A815 S31803 WP-SWP-WX, ASME B16.9,	PCS	1	0	1	
49	90 DEG. LR. ELBOW	D07	2"	--	80S	--	1500#	ASME B16.9	ASTM A815 S31803 WP-SWP-WX, ASME B16.9,	PCS	5	1	6	
50	90 DEG. LR. ELBOW	C06	36"	--	38.2mm	--	900#	ASME B16.10	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	9	1	10	
51	90 DEG. LR. ELBOW	C06	24"	--	100	--	900#	ASME B16.9	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	3	0	3	
52	90 DEG. LR. ELBOW	C06	12"	--	140	--	900#	ASME B16.9	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	10	1	11	
53	90 DEG. LR. ELBOW	C06	8"	--	140	--	900#	ASME B16.9	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	8	1	9	
54	90 DEG. LR. ELBOW	C06	6"	--	160	--	900#	ASME B16.9	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	32	3	35	
55	90 DEG. LR. ELBOW	C06	4"	--	XXS	--	900#	ASME B16.9	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	31	3	34	
56	90 DEG. LR. ELBOW	C06	2"	--	XXS	--	900#	ASME B16.9	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	32	3	35	
57	90 DEG. LR. ELBOW	C06	3/4"	--	XXS	--	900#	ASME B16.9	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	11	1	12	
58	90 DEG. LR. ELBOW	B06	6"	--	XS	--	150#	ASME B16.9	'ASTM A234 GR. WPB; SMLS; BW; ASME B16.9	PCS	10	1	11	
59	90 DEG. LR. ELBOW	B06	4"	--	XXS	--	150#	ASME B16.10	'ASTM A234 GR. WPB; SMLS; BW; ASME B16.9	PCS	17	2	19	
60	90 DEG. LR. ELBOW	B06	2"	--	XXS	--	150#	ASME B16.9	ASTM A234 GR. WPB; SMLS; BW; ASME B16.9	PCS	6	1	7	
FLANGES														
61	FLANGE	D70	36"	--	59.2mm	--	1500#	ASME B16.5	ASTM A182 F51, 'WN, 1500#, RTJ, ASME B16.47 SERIES A	PCS	5	1	6	
62	BLIND FLANGE	D70	36"	--	59.2mm	--	1500#	ASME B16.5	ASTM A182 F51, 1500#, RTJ, ASME B16.47 SERIES A	PCS	1	0	1	
63	FLANGE	C06	36"	--	38.2mm	--	900#	ASME B16.5	ASTM A694 F60, 'WN, 900#, RTJ, ASME B16.47 SERIES A	PCS	23	2	25	
64	BLIND FLANGE	C06	36"	--	38.2mm	--	900#	ASME B16.5	ASTM A694 F60, 900#, RTJ, ASME B16.47 SERIES A	PCS	1	0	1	
65	FLANGE	C06	24"	--	100	--	900#	ASME B16.5	ASTM A694 F60, 'WN, 900#, RTJ, ASME B16.5	PCS	4	0	4	
66	FLANGE	C06	12"	--	140	--	900#	ASME B16.5	ASTM A105N; WN; 900#, RTJ; ASME B16.5	PCS	6	1	7	
67	FLANGE	C06	8"	--	140	--	900#	ASME B16.5	ASTM A105N; WN; 900#, RTJ; ASME B16.5	PCS	3	0	3	
68	FLANGE	C06	6"	--	160	--	900#	ASME B16.5	ASTM A105N; WN; 900#, RTJ; ASME B16.5	PCS	17	2	19	
69	FLANGE	C06	4"	-	XXS	-	900#	ASME B16.5	ASTM A105N; WN; 900#, RTJ; ASME B16.5	PCS	51	5	56	
70	FLANGE	C06	2"	--	XXS	--	900#	ASME B16.5	ASTM A105N; WN; 900#, RTJ; ASME B16.5	PCS	50	5	55	
71	FLANGE	C06	3/4"	--	XXS	--	1500#	ASME B16.5	ASTM A105N; SW; 1500#, RTJ; ASME B16.5	PCS	10	1	11	
72	BLIND FLANGE	C06	3/4"	--	XXS	--	1500#	ASME B16.5	ASTM A105N; SW; 1500#, RTJ; ASME B16.5	PCS	10	1	11	
73	FLANGE	B06	6"	--	XS	--	150#	ASME B16.5	ASTM A105N; WN; 150#, RF; ASME B16.5	PCS	19	2	21	
74	BLIND FLANGE	B06	6"	--	XS	--	150#	ASME B16.5	'ASTM A105N; WN; 150#, RF; ASME B16.5	PCS	1	0	1	
75	FLANGE	B06	3/4"	--	XXS	--	150#	ASME B16.5	'ASTM A105N; SW; 150#, RF; ASME B16.5	PCS	5	1	6	
76	BLIND FLANGE	B06	3/4"	--	XXS	--	150#	ASME B16.5	'ASTM A105N; SW; 150#, RF; ASME B16.5	PCS	5	1	6	

SPECTACLE BLINDS														
77	SPECTACLE BLIND	C06	4"	--	--	--	900#	ASME B16.48	ASTM A516 GR.70; 900#; ASME B16.48	PCS	6	1	7	
GASKETS														
78	GASKET	D70	36"	--	--	--	1500#	ASME B16.20	RING JOINT, OCTAGONAL, CL1500, ASME B16.20, TYPE 316	PCS	3	3	6	
79	GASKET	D70	2"	--	--	--	1500#	ASME B16.20	RING JOINT, OCTAGONAL, CL1500, ASME B16.20, TYPE 317	PCS	3	3	6	
80	GASKET	C06	36"	--	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	18	18	36	
81	GASKET	C06	24"	--	--	--	900#	'ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	5	5	10	
82	GASKET	C06	12"	--	--	--	900#	'ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	8	8	16	
83	GASKET	C06	8"	--	--	--	900#	'ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	4	4	8	
84	GASKET	C06	6"	--	--	--	900#	'ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	18	18	36	
85	GASKET	C06	4"	--	--	--	900#	'ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	59	59	118	
86	GASKET	C06	2"	--	--	--	900#	'ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	17	17	34	
87	GASKET	C06	1 1/2"	--	--	--	900#	'ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	2	2	4	
88	GASKET	C06	3/4"	--	--	--	900#	'ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	10	10	20	
89	GASKET	B06	6"	--	--	--	150#	ASME B16.20	SPIRAL WOUND, LOW STRESS, 150#, ASME B16.20, SS 316 WINDING 4.5MM THICK, GRAPHITE FILLER, INNER AND CENTERING RING 3.2 MM THICK	PCS	21	21	42	
90	GASKET	B06	2"	--	--	--	150#	ASME B16.20	SPIRAL WOUND, LOW STRESS, 150#, ASME B16.20, SS 316 WINDING 4.5MM THICK, GRAPHITE FILLER, INNER AND CENTERING RING 3.2 MM THICK	PCS	7	7	14	
91	GASKET	B06	3/4"	--	--	--	150#	ASME B16.20	SPIRAL WOUND, LOW STRESS, 150#, ASME B16.20, SS 316 WINDING 4.5MM THICK, GRAPHITE FILLER, INNER AND CENTERING RING 3.2 MM THICK	PCS	5	5	10	
STUDBOLTS & NUTS														
92	STUD BOLTS & HEX NUTS	D70	3+1/2"	640mm	--	--	1500#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A453 GRADE 660 CLASS B	PCS	60	12	72	
93	STUD BOLTS & HEX NUTS	D70	7/8"	160mm	--	--	1500#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A453 GRADE 660 CLASS B	PCS	24	5	29	
94	STUD BOLTS & HEX NUTS	D06	3+1/2"	640mm	--	--	1500#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A155	PCS	24	5	29	
95	STUD BOLTS & HEX NUTS	C06	3+1/2"	605mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A155	PCS	360	72	432	
96	STUD BOLTS & HEX NUTS	C06	2+1/2"	505mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	100	20	120	
97	STUD BOLTS & HEX NUTS	C06	1+3/8"	280mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	160	32	192	
98	STUD BOLTS & HEX NUTS	C06	1+3/8"	230mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	48	10	58	
99	STUD BOLTS & HEX NUTS	C06	1+1/8"	215mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	216	43	259	
100	STUD BOLTS & HEX NUTS	C06	1+1/8"	190mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	472	94	566	
101	STUD BOLTS & HEX NUTS	C06	7/8"	160mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	118	24	142	
102	STUD BOLTS & HEX NUTS	C06	7/8"	140mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	8	2	10	
103	STUD BOLTS & HEX NUTS	C06	3/4"	125mm	--	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	32	6	38	
104	STUD BOLTS & HEX NUTS	B06	3/4"	105mm	--	--	150#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	168	34	202	
105	STUD BOLTS & HEX NUTS	B06	5/8"	95mm	--	--	150#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	16	3	19	
106	STUD BOLTS & HEX NUTS	B06	1/2"	75mm	--	--	150#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A154	PCS	4	1	5	
107	STUD BOLTS & HEX NUTS	B06	1/2"	65mm	--	--	150#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	40	8	48	
SPECIAL PIPING ITEMS														
108	QUICK DISCONNECT COUPLING	C06	2"	--	--	--	900#	ASTM F1122	BODY: ASTM A105N; CAP: ASTM A234 GR. WPB; HOSE: NITRILE / EPDM RUBBER WITH HT SYNTHETIC REINFORCEMENT; FLANGE: ASTM A105N; SEAL: RPTFE	1 SET	3	0	3	

2.0 MATERIAL TAKE OFF FOR PIPING VALVES - ELEPA														COMMENTS
S/NO	SIZE	MATERIAL CLASS	VALVE TYPE	VALVE CODE	BORE TYPE	END CONNECTION	PRESSURE CLASS	DESIGN CODE	ITEM & MATERIAL DESCRIPTION	UNIT	REQUIRED QUANTITY	CONTINGENCY	TOTAL QUANTITY TO PROCURE	
109	36" SDV 1001	D06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1	Material is captured in LLI List
110	36"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	4	0	4	Material is captured in LLI List
111	36"	C06	BALL	—	FB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	2	0	2	Material is captured in LLI List
112	24"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	2	0	2	Material is captured in LLI List
113	12"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	4	0	4	Material is captured in LLI List
114	8"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	2	0	2	Material is captured in LLI List
115	6"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	7	0	7	Material is captured in LLI List
116	6"	B06	BALL	—	FB	RF FLANGED WITH SMOOTH FINISH	150#		FLGD, RF, CL150, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	7	0	7	Material is captured in LLI List
117	4" BDV 1001	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1	Material is captured in LLI List
118	4"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	18	0	18	Material is captured in LLI List
119	2"	D70	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	1500#		FLGD, RJ, CL1500, API SPEC 6D, ASTM A182 F51 OR ASTM A995 GR 4A BODY, UNS S31803 TRIM, SPB/TPE, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1	Material is captured in LLI List
120	2"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	24	0	24	Material is captured in LLI List
121	2"	B06	BALL	—	RB	RF FLANGED WITH SMOOTH FINISH	150#		FLGD, RF, CL150, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, FLOATINF BALL, LEVER OPR .	PCS	3	0	3	Material is captured in LLI List
122	3/4"	C06	BALL	—		SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	1500#		SW, CL1500, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	10	0	10	Material is captured in LLI List
123	3/4"	B06	BALL	—		SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	800#		SW, CL800, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	5	0	5	Material is captured in LLI List
124	4"	C06	GLOBE	—		RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SW/VEL PLUG DISC, HNDWHL OPR W/ GEARBOX	PCS	13	0	13	Material is captured in LLI List
125	2"	CO6	GLOBE	—		RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SW/VEL PLUG DISC, HNDWHL OPR W/ GEARBOX	PCS	9	0	9	Material is captured in LLI List
126	2"	CO6	BALL	—		RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	12	0	12	Material is captured in LLI List

1. MATERIAL TAKE OFF FOR PIPING BULKS - AKABEL RECEIVER														COMMENTS
S/NO	ITEM DESCRIPTION	MATERIAL CLASS	SIZE 1	SIZE 2	SCHEDULE / THICKNESS 1	SCHEDULE / THICKNESS 2	PRESSURE RATING	DESIGN CODE	MATERIAL DESCRIPTION	UNIT	QUANTITY REQUIRED	CONTINGENCY	QUANTITY TO PROCURE	
PIPES														
1	PIPE	C06	36"	_	38.2mm	--	900#	ASME B36.10M	WELDED, SAW 100% RADIOGRAPHY, ASME B36.10, API SPEC 5L GRADE X60 PSL 2 (12m SRL)	m	16.7	7.3	24	Material is captured in LLI List
2	PIPE	C06	12"	_	140	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	15.8	8.2	24	Material is captured in LLI List
3	PIPE	C06	4"	_	XXS	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (6m SRL)	m	37.4	34.6	72	Material is captured in LLI List
4	PIPE	C06	2"	_	XXS	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (6m SRL)	m	9.4	20.6	30	Material is captured in LLI List
5	PIPE	C06	3/4"	_	XXS	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (6m SRL)	m	0.3	5.7	6	Material is captured in LLI List
6	PIPE	B06	6"	_	XS	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (6m SRL)	m	28.8	7.2	36	Material is captured in LLI List
7	PIPE	B06	4"	_	120	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (6m SRL)	m	6.2	41.8	48	Material is captured in LLI List
8	PIPE	B06	3/4"	_	XXS	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (6m SRL)	m	0.1	5.9	6	Material is captured in LLI List
TEES														
9	REDUCING TEE	C06	36"	12"	38.2mm	140	900#	ASME B16.9	ASTM A860 WPHY 60, BW, ASME B16.9	PCS	2	0	2	
10	EQUAL TEE	C06	4"	4"	XXS	XXS	900#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	5	1	6	
11	EQUAL TEE	C06	2"	2"	XXS	XXS	900#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	1	0	1	
12	EQUAL TEE	B06	6"	6"	XS	XS	150#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	2	0	2	
13	REDUCING TEE	B06	6"	4"	XS	140	150#	ASME B16.9	ASTM A234 WPB, BW, ASME B16.9	PCS	1	0	1	
WELDOLETS														
14	WELDOLET	C06	36"	2"	38.2mm	XXS	900#	MSS-SP-97	WELDOLET, MSS SP-97, ASTM A105N	PCS	1	0	1	
15	WELDOLET	C06	12"	4"	140	XXS	900#	MSS-SP-97	WELDOLET, MSS SP-97, ASTM A105N	PCS	1	0	1	
16	WELDOLET	C06	12"	2"	140	XXS	900#	MSS-SP-97	WELDOLET, MSS SP-97, ASTM A105N	PCS	2	0	2	
SOCKOLETS														
17	SOCKOLET	C06	12"	3/4"	140	XXS	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	1	0	1	
18	SOCKOLET	C06	4"	3/4"	XXS	XXS	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	1	0	1	
19	SOCKOLET	B06	6"	3/4"	XS	XXS	6000#	MSS-SP-97	SOCKOLET, SW, ASME B16.11, MSS SP-97, A105N	PCS	1	0	1	
REDUCERS														
20	CONCENTRIC REDUCER	C06	36"	18"	38.2mm	140	900#	ASME B16.9	ASTM A234 WPB, BW, WPB/ASME B16.9,	PCS	1	0	1	
21	CONCENTRIC REDUCER	B06	6"	4"	XS	XXS	150#	ASME B16.9	ASTM A234 GR. WPB; SMLS; BW; ASME B16.8	PCS	1	0	1	
22	CONCENTRIC REDUCER	B06	4"	2"	XXS	XXS	150#	ASME B16.9	ASTM A234 GR. WPB; SMLS; BW; ASME B16.9	PCS	1	0	1	
ELBOW														
23	90 DEG. LR. ELBOW	C06	12"	_	140	--	900#	ASME B16.9	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	3	0	3	
24	90 DEG. LR. ELBOW	C06	4"	_	XXS	--	900#	ASME B16.9	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	11	1	12	
25	90 DEG. LR. ELBOW	C06	2"	_	XXS	--	900#	ASME B16.9	ASTM A860 WPHY 60; BW; ASME B16.9	PCS	9	1	10	
26	90 DEG. LR. ELBOW	B06	6"	_	XS	--	150#	ASME B16.9	ASTM A234 GR. WPB; SMLS; BW; ASME B16.9	PCS	3	0	3	
27	90 DEG. LR. ELBOW	B06	4"	_	XXS	--	150#	ASME B16.10	ASTM A234 GR. WPB; SMLS; BW; ASME B16.9	PCS	4	0	4	
28	90 DEG. LR. ELBOW	B06	2"	_	XXS	--	150#	ASME B16.9	ASTM A234 GR. WPB; SMLS; BW; ASME B16.9	PCS	6	1	7	
FLANGES														
29	FLANGE	C06	36"	_	38.2mm	--	900#	ASME B16.5	ASTM A694 F60, WN, 900#, RTJ, ASME B16.47 SERIES A	PCS	3	0	3	
30	BLIND FLANGE	C06	36"	_	38.2mm	--	900#	ASME B16.5	ASTM A694 F60, 900#, RTJ, ASME B16.47 SERIES A	PCS	1	0	1	
31	FLANGE	C06	12"	_	140	--	900#	ASME B16.5	ASTM A105N; WN; 900#; RTJ; ASME B16.5	PCS	3	0	3	

32	FLANGE	C06	4"	—	XXS	-	900#	ASME B16.5	ASTM A105N; WN; 900#; RTJ; ASME B16.5	PCS	14	1	15	
33	FLANGE	C06	2"	—	XXS	--	900#	ASME B16.5	ASTM A105N; WN; 900#; RTJ; ASME B16.5	PCS	17	2	19	
34	FLANGE	C06	3/4"	—	XXS	--	1500#	ASME B16.5	ASTM A105N; SW; 1500#; RTJ; ASME B16.5	PCS	2	0	2	
35	BLIND FLANGE	C06	3/4"	—	XXS	--	1500#	ASME B16.5	ASTM A105N; SW; 1500#; RTJ; ASME B16.5	PCS	2	0	2	
36	FLANGE	B06	6"	—	XS	--	150#	ASME B16.5	ASTM A105N; WN; 150#; RF; ASME B16.5	PCS	6	1	7	
37	BLIND FLANGE	B06	6"	—	XS	--	150#	ASME B16.5	ASTM A105N; WN; 150#; RF; ASME B16.5	PCS	1	0	1	
38	FLANGE	B06	3/4"	—	XXS	--	150#	ASME B16.5	ASTM A105N; SW; 150#; RF; ASME B16.5	PCS	1	0	1	
39	BLIND FLANGE	B06	3/4"	—	XXS	--	150#	ASME B16.5	ASTM A105N; SW; 150#; RF; ASME B16.5	PCS	1	0	1	
SPECTACLE BLINDS														
40	SPECTACLE BLIND	C06	4"	—	--	--	900#	ASME B16.48	ASTM A516 GR.70; 900#; ASME B16.48	PCS	1	0	1	
GASKETS														
41	GASKET	C06	36"	—	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	3	3	6	
42	GASKET	C06	12"	—	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	4	4	8	
43	GASKET	C06	4"	—	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	16	16	32	
44	GASKET	C06	2"	—	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	3	3	6	
45	GASKET	C06	3/4"	—	--	--	900#	ASME B16.20	RING JOINT, OCTAGONAL, 900#, ASME B16.20, SOFT IRON	PCS	4	4	8	
46	GASKET	B06	6"	—	--	--	150#	ASME B16.20	SPIRAL WOUND, LOW STRESS, 150#, ASME B16.20, SS 316 WINDING 4.5MM THICK, GRAPHITE FILLER, INNER AND CENTERING RING 3.2 MM THICK	PCS	5	5	10	
47	GASKET	B06	3/4"	—	--	--	150#	ASME B16.20	SPIRAL WOUND, LOW STRESS, 150#, ASME B16.20, SS 316 WINDING 4.5MM THICK, GRAPHITE FILLER, INNER AND CENTERING RING 3.2 MM THICK	PCS	1	1	2	
STUDBOLTS & NUTS														
48	STUD BOLTS & HEX NUTS	C06	3+1/2"	—	640mm	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A155	PCS	60	12	72	
49	STUD BOLTS & HEX NUTS	C06	1+3/8"	—	280mm	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	80	16	96	
50	STUD BOLTS & HEX NUTS	C06	1+1/8"	—	280mm	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	128	26	154	
51	STUD BOLTS & HEX NUTS	C06	7/8"	—	190mm	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	24	5	29	
52	STUD BOLTS & HEX NUTS	C06	3/4"	—	125mm	--	900#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	16	3	19	
53	STUD BOLTS & HEX NUTS	B06	5/8"	—	105mm	--	150#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153	PCS	40	8	48	
54	STUD BOLTS & HEX NUTS	B06	1/2"	—	65mm	--	150#	ASME B18.2.1 ASME B18.2.2	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A154	PCS	4	1	5	
SPECIAL PIPING ITEMS														
55	QUICK DISCONNECT COUPLING	C06	2"	—	--	--	900#	ASTM F1122	BODY: ASTM A105N; CAP: ASTM A234 GR. WPB; HOSE: NITRILE / EPDM RUBBER WITH HT SYNTHETIC REINFORCEMENT; FLANGE: ASTM A105N; SEAL: RPTFE	1 SET	3	0	3	

2.0 MATERIAL TAKE OFF FOR PIPING VALVES -AKABEL RECEIVER														COMMENTS
S/NO	SIZE	MATERIAL CLASS	VALVE TYPE	VALVE CODE	BORE TYPE	END CONNECTION	PRESSURE CLASS	DESIGN CODE	ITEM & MATERIAL DESCRIPTION	UNIT	REQUIRED QUANTITY	CONTINGENCY	TOTAL QUANTITY TO	
1	36"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1	Material is captured in LLI List
3	12"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	2	0	2	Material is captured in LLI List
6	6"	B06	BALL	—	FB	RF FLANGED WITH SMOOTH FINISH	150#		FLGD, RF, CL150, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	2	0	2	Material is captured in LLI List
7	4"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	5	0	5	Material is captured in LLI List
9	2"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	7	0	7	Material is captured in LLI List
11	3/4"	C06	BALL	—	—	SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	1500#		SW, CL1500, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	2	0	2	Material is captured in LLI List
12	3/4"	B06	BALL	—	—	SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	800#	—	SW, CL800, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	2	0	2	Material is captured in LLI List
13	4"	C06	GLOBE	—	—	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SWIVEL PLUG DISC, HNDWHL OPR W/ GEARBOX	PCS	2	0	2	Material is captured in LLI List
14	2"	CO6	GLOBE	—	—	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SWIVEL PLUG DISC, HNDWHL OPR W/ GEARBOX	PCS	3	0	3	Material is captured in LLI List
15	2"	CO6	DOUBLE BLOCK & BLEED	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	6	0	6	Material is captured in LLI List